

AP-2018
Grade-10
General science paper-1
Parts A and B

[Time: 2 hrs. 45 min]

[Maximum Marks: 40]

Instructions:

1. This paper contains Part-A
2. Answer the questions. under Part-A on the separate answer book. Answer all the questions: 'Internal choice is given to the questions under Section-III.
3. In the duration, of 2.45 hours, 15 minutes of time is allotted to read the question paper.

Part-A

Time: 2 hours

Marks: 30

Instructions:

1. Part-A comprises three sections I, II and III.
2. All the questions are compulsory.
3. There is no overall choice. However, there is an internal choice to the questions under Section-III.

SECTION – 1

4x1=4

1. Iron gets rust: but Gold doesn't, why?

Ans: Gold doesn't react with the oxygen but on the other hand, iron does. This is the only reason why the iron gets rust, but gold doesn't.

2. State Fermat's principle.

Ans: In a Fermat's principle light chooses the path which takes the least time to travel. It is also applicable to the reflection of light.

3. Can a virtual image be photographed by a camera?

Ans: Yes, the virtual image can be formed by a camera. The secondary image of the image created by the camera is called the virtual image.

4. Give the names of the functional groups.

(a) -COOR

Ans: a. - COOR - The functional group present in an ester is - COOR The IUPAC group suffix of an ester is -ate. Example: Ethyl acetate.

(a) -OH

Ans: b. - OH - hydroxyl group

SECTION – II

Note:**(5 x2 = 10)**

- (i) Answer all the questions
- (ii) Answer each question in 4 or 5 sentences
- (iii) Each question carries 2 marks

5. Your friend has a doubt that whether a concave mirror or a convex mirror is used as a rear - view mirror in the vehicles. What questions will you ask to clarify his doubts?

Ans: Convex mirror has a much wider field of view and gives an erect, virtual, full sized diminished image of distant object, whereas a concave mirror forms a real, erect, virtual image and the size can be smaller. Therefore, the convex mirror is used as a rear side view mirror in the vehicle.

6. The focal length of a converging lens is 20 cm. Where will the image be formed, if an object is placed at 60 cm from the lens? Write characteristics of the image.

Ans: Focal length (F) = 20cm

object distance (u) = -60cm

Image distance (v) = ?

Using Lens formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$\frac{1}{20} = \frac{1}{v} - \left(\frac{1}{-60} \right)$$

$$\frac{1}{20} = \frac{1}{v} + \frac{1}{60}$$

$$\frac{1}{20} - \frac{1}{60} = \frac{1}{v}$$

$$\frac{3-1}{\frac{60}{2}} = \frac{1}{v}$$

$$\frac{2}{60} = \frac{1}{v}$$

$$V = 60/2 = 30\text{cm}$$

As the object is beyond the Centre of curvature ($2F = 40\text{cm}$), then the image will be formed between focus & Centre of curvature. The image will be real, diminished, inverted and 30 cm away from the lens.

7. **What is the reason for the blue colour of the sky? How do you appreciate the role of molecules in the atmosphere in this regard?**

Ans: The sky appears to be blue because of the refraction and scattering of the light through the molecules. In the atmosphere due to the mixture of dust particles, smoke particles, molecules of air, water droplets, etc. the light gets scattered and the sky appears to be blue.

8. **Name the principles which says an Orbital can hold only 2 electrons and explain.**

Ans: According to Pauli's exclusion principle no two electrons can have same four electronic quantum number. So according to this an orbital can hold only 2 electrons that too having opposite spins.

9. **How do you appreciate the special nature of Inert gases?**

Ans: The inert gases are colorless, odorless, tasteless, and non-flammable gases

SECTION –III

NOTE:

- (i) Answer all the questions. | $4 \times 4 = 16$
- (ii) Answer each question in 8-10 sentences.
- (iii) There is internal choice for each question.
- (iv) Only one option from each question is to be attempted.
- (v) Each question carries FOUR marks.

10. Answer the following questions by using the data given in the table.

Substance	Specific Heat (Cal/g°C)
Lead	0.031
Aluminium	0.21
Copper	0.095
Water	1.00
Iron	0.115

a) Write SI units for Specific Heat.

Ans: The SI units of specific heats are J/kgK (kJ/kg°C).

b) Based on Specific Heat values arrange the substances given in the table in ascending order.

Ans: Lead < Copper < Aluminum < Water

c) If we supply same quantity of heat, which substance will heat up faster?

Ans: copper

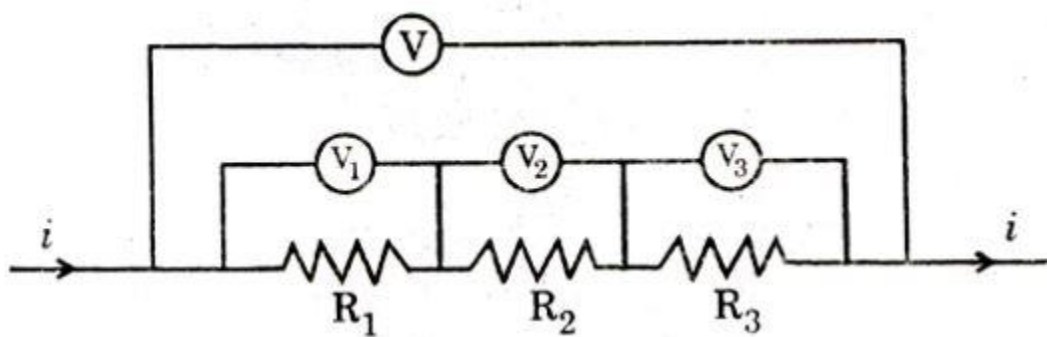
d) Calculate the amount of heat required to raise the

temperature of 1 kg of Iron through 10°C .

Ans: Heat (J) = mass of substance (g) $\times$ heat capacity of substance (J/g.K) $\times$ the change in temperature (K).

In this case the heat required = $1000 \times 0.45 \times 10 = 4500 \text{ J}$ or 4.5 kJ

OR



Observe the above diagram and answer the following.

(a) Are all the resistors connected in parallel or series?

Ans. all resistors are connected in series

(b) What is the equivalent resistance of the combination of three resistors?

Resistance in series

$$R = R_1 + R_2 + R_3$$

(c) In this system which physical quantity is constant?

In this system "I" is constant.

(d) If $R_1 = 2\Omega$, $R_2 = 3\Omega$, $R_3 = 4\Omega$, find equivalent resistance.

$$R = R_1 + R_2 + R_3 = 2\Omega + 3\Omega + 4\Omega = 9\Omega$$

11. Write the chemical formulae for Washing soda and Baking soda and give their uses.

Ans:

Washing Soda - Na_2CO_3

Baking Soda - NaHCO_3

Washing Soda	Baking soda
• Used in glass, soap and paper industries. • It is used for removing the permanent hardness of water	• Used as mild antiseptic. • It is used in soda – acid fire extinguishers

OR

Who proposed the Valence Bond Theory? Explain the formation of N_2 molecule by using this theory.

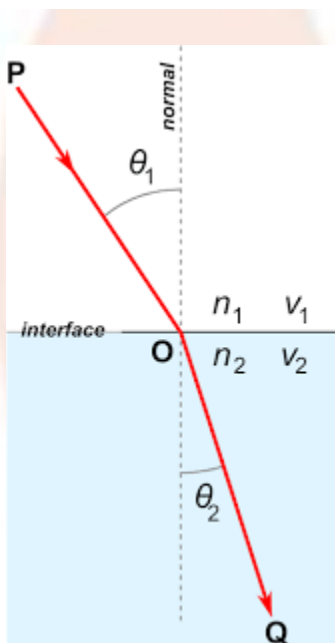
Ans: The Valence bond theory was proposed by the American scientists Linus Pauling and John C. Slater.

The formation of N_2 molecules by valence bond theory.

- The electronic configuration of 'N' atom is 2, 5 and to have Octet in the valence shell it requires three more electrons.
- When two nitrogen atoms approach each other, each atom contributes 3 electrons for bonding.
- There are six electrons shared between two nitrogen atoms in the form of three pairs.
- Therefore, there is a triple bond between two nitrogen atoms in N_2 molecules.

12. Explain the relation between angle of incidence and angle of refraction with an experiment.

Ans: When the light ray passes to a denser medium from a less dense medium then its speed decreases, the ray is deflected towards the normal; the angle of incidence is greater than the angle of refraction. The ratio between the angle of incidence and the angle of refraction is inversely proportional to the ratio of the refractive indices of the two media.



OR

Write an activity to each of the following chemical reaction.

[A] Photochemical reaction.

Ans: Conversion of sunlight into chemical energy is known as photochemical reaction. Carbohydrates and Oxygen are the bi-product formed and is very essential for life.

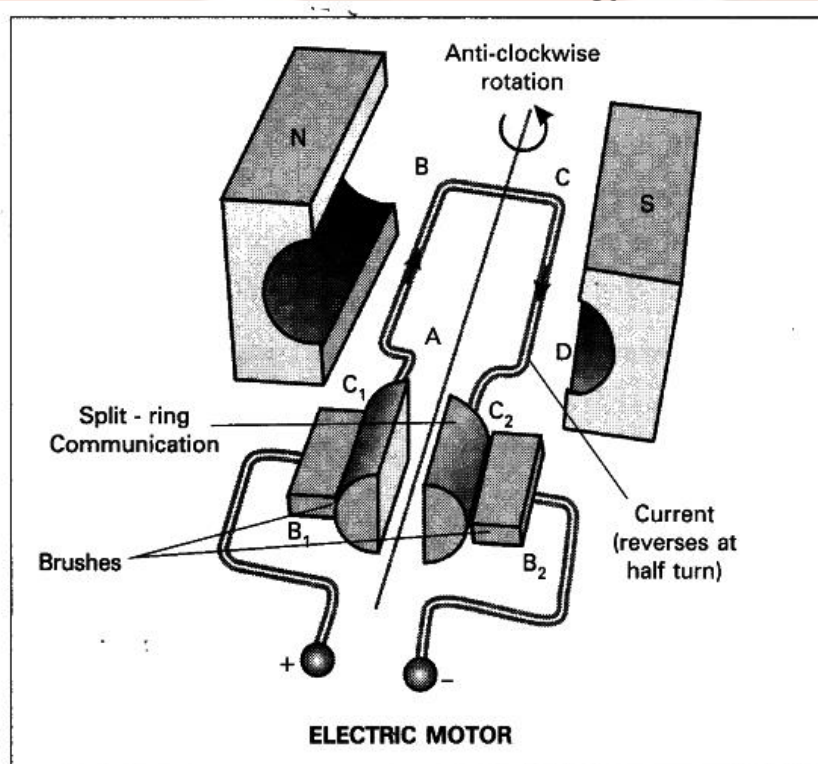
(B). Chemical displacement reaction.

Ans: When a more relative metal replaces a less reactive metal from its compound it is defined as chemical displacement reaction.

13. Name the device that converts electrical energy into mechanical energy. Draw its diagram and label the parts.

Ans: A motor converts electrical energy into mechanical

energy.

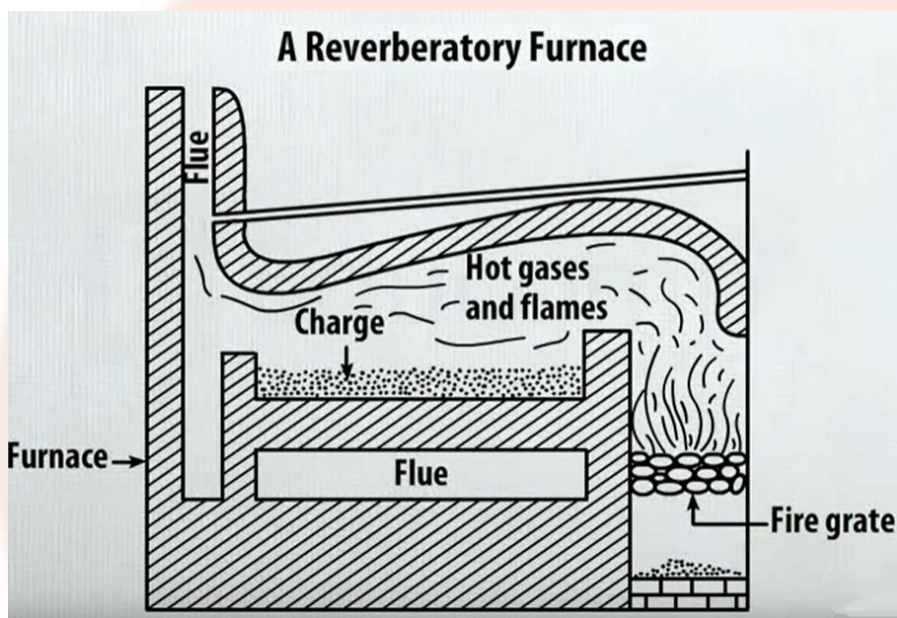


OR

What is a Furnace? Draw Reverberatory furnace and label its parts.

Ans: A furnace is used to extract metals from their ores. There are different types of furnaces like the Siemens martin open-hearth furnace and the Bessemer converter.

Reverberatory furnace:



Part – B**SECTION – IV****Time: 30min****Marks: 10**

$$20 \times \frac{1}{2} = 10$$

- 1, Answer all the questions.
2. Each question carries 1/2 marks
3. Marks will not be awarded in any case of over-writing, rewritten or erased answers.
4. Write the CAPITAL LETTER (A, B, C, D) showing the correct answer for the following questions in the brackets provided against them.

14. The SI. unit of Heat is []

- (A) Calorie
- (B) Joule
- (C) Calorie / g°C
- (D) Joule / Kg-kelvin

Ans (B). Joule

The S.I. unit of Heat is Joule.

15. $x \text{ KClO}_3 \rightarrow y \text{ KCl} + z \text{ O}_2$. The respective values of x, y, z are []

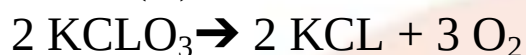
- (A) 1,2,3

(B) 3, 3, 2

(C) 2, 2, 3

(D) 2, 2, 2

Ans: (C). 2, 2, 3



16. The iron nail dipped in Copper sulphate solution becomes brown and the blue colour of the Copper sulphate solution fades.

Which type of reaction is this ?

(A) Chemical combination. _

(B) Chemical decomposition.

(C) Double decomposition.

(D) Displacement.

Ans: (D). The iron nail become brownish in colour and the blue colour of copper sulphate solution fade because when an iron nail dipped in the copper sulphate solution then iron displaces copper from the copper sulphate because iron is more reactive than copper therefore the colour of copper sulphate solution changes.

The reaction involved is - $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$

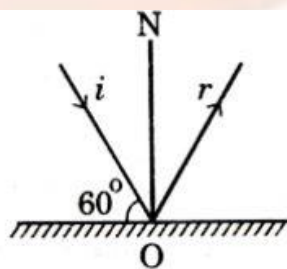
17. Find the values of $\angle i$, $\angle r$

(A) $\angle i = 60^\circ$, $\angle r = 60^\circ$

(B) $\angle i = 60^\circ$, $\angle r = 30^\circ$

(C) $\angle i = 30^\circ$, $\angle r = 60^\circ$

(D) $\angle i = 30^\circ$, $\angle r = 30^\circ$



in the diagram.

Ans: (D) { since ,Angle of incidence =Angle of refraction ,so both angle will be 30 degree }

18.The colour of Methyl orange in alkali solutions is

- (A) Yellow
- (B) Orange
- (C) Red
- (D) Blue

Ans: (A). In a solution becoming less acidic, methyl orange moves from red to orange and finally to yellow with the reverse occurring for a solution increasing in acidity. The entire color change occurs in acidic conditions. In an acid, it is reddish and in alkali, it is yellow.

19. Which of the following is Snell's law?

(A) $n_1 \sin i = \frac{n_2}{\sin r}$

(B) $\frac{n_1}{n_2} = \frac{\sin i}{\sin r}$

(C) $\frac{n_2}{n_1} = \frac{\sin i}{\sin r}$

(D) $\frac{n_1}{\sin i} = n_2 \sin r$

Ans: Ans: (A). Snells law explains the relation between the angle of incidence and the angle of refraction. It can be shown

$$n_1 \sin i = n_2 \sin r$$

20. Where should be the object placed on the principle axis of a Convex lens in order to get virtual image?

- (A) Between Optic centre and F.
- (B) At F
- (C) Between F and C.
- (D) At C

Ans: (A) Converging lens produced a virtual image when the object is placed in front of the focal point. For such a position, the image is magnified and upright, thus allowing for easier viewing.

21. Statement P : Myopia can be corrected by using a bi-concave lens.

Statement Q : For a bi-concave lens, f is positive.

- (A) P is false, Q is true
- (B) P is true, Q is false.
- (C) Both P and Q are true.
- (D) Both P and Q are false.

Ans: Option D

- Myopia can be corrected using concave lens not by bi-concave lens so statement P is false.
- For bi-concave lens f is always taken as negative so statement Q is also false.

22. The splitting of white light into 7 colours is called

- (A) Scattering

- (B) Reflection
- (C) Refraction
- (D) Dispersion

Ans: A

When a white light splits into 7 colours is called as scattering. It was discovered when the light was passed via a prism and it got scattered into 7 different colors.

23. Which rule is violated in the electronic configuration $1s^2 2s^0 2s^2$?

- (A) Aufbau principle.
- (B) Hand's rule.
- (C) Pauli exclusion Principle
- (D) Octet rule.

Ans: (A). Aufbau principle is violated in this electronic configuration because according to Aufbau principle, Electron enters orbital of lower energy.

24. If $n = 2$, then angular momentum quantum number l values =

- (A) 0,1
- (B) 0,1, 2
- (C) 0
- (D) 1,2

Ans: (A). The number of values of the orbital angular number l can also be used to identify the number of subshells in a principal electron shell:

25. On moving from top to bottom in a group, the values of Ionization energy

- (A) Increases.
- (B) Decreases.
- (C) No change.
- (D) Can't say.

Ans: (B).

Due to electron shielding the value of ionization energy of the elements within a group generally decreases from top to bottom

26. The bond present in HCL molecule is

- (A) Ionic bond.
- (B) Polar covalent bond.
- (C) Non-polar covalent bond.
- (D) None.

Ans: (B).

HCL has polar covalent bond. Since atoms with different electron negativities share electrons in a covalent bond.

27. Shape of Ammonia molecule is

- (A) Linear
- (B) Trigonal - Plana
- (C) Tetrahedron
- (D) Trigonal Pyramid

Ans: (D).

The ammonia molecule has a trigonal pyramidal shape. The bond angle is 106.7° according to VSEPR theory.

The central nitrogen atom has five outer electrons with an additional electron from each hydrogen atom.

28. An unknown circuit draws a current of 2A from a 12V battery. Its equivalent resistance is

- (A) $240\ \Omega$
- (B) $6\ \Omega$
- (C) $12\ \Omega$
- (D) $2\ \Omega$

Ans: (D). Current = 2A

Voltage = 12V

We know $V = IR$

$$R = V/I = 12/2 = 2$$

R = $6\ \Omega$

29. The main difference between A.C. generator and D.C. generator is.....

- (A) Carbon brushes
- (B) Magnets
- (C) Coil
- (D) Commutator

Ans: (C).

AC generators current changes its directions periodically so coil remains still while the magnet associated with it revolves. Whereas in DC generator the current flows uni-directional.

30. Observe the following table.

Metal	Ore
P Mercury R	Bauxite Q Hematite

Identify the substances which are to be present in P Q, R positions.

- (A) Aluminum, Cinnabar, Iron.
- (B) Sodium, Galena, Magnesium.
- (C) Sodium, Cinnabar, Iron.
- (D) Magnesium, Galena, Iron.

Ans: (A).

- Bauxite is the principal ore of Aluminum.
- Cinnabar is the ore of Mercury
- Hematite is the ore of Iron

31. When Acetic acid reacts with Ethyl alcohol, we add conc. H_2SO_4 .

This process is called

- (A) Saponification.
- (B) Esterification.
- (C) Catenation
- (D) Isomerism

Ans: (B).

The reaction is termed as Esterification reaction. Sulphuric acid act as an catalyst and it is a slow and reversible reaction.

32. Graphite and Diamonds are two

- (A) Isomers
- (B) Allotropes
- (C) Homologs
- (D) Metals

Ans: (B). Diamond and graphite are two allotropes of carbon: pure forms of the same element that differ in structure.

33. The name of $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{COOH}$ is

- (A) Propanoic acid (B) Propanaldehyde
- (C) Butanoic acid (D) Butanaldehyde

Ans: (C). The IUPAC name of $\text{CH}_3.\text{CH}_2.\text{CH}_2\text{COOH}$ is butanoic acid. As there are 3 -C, so it is but and there is single bond so it's ane. But for COOH it'll be oic acid.